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Paleomagnetism of the Lower Cretaceous redbeds from northern Spain: evidence for a multistage acquisition of magnetization

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Abstract

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An extensive study of the Wealden series from the Cantabric Chain and the Sierra de los Cameros (northern Spain) reveals that the N.R.M. contains four types of components tentatively classified in decreasing ages. The older components (group 1) are considered as the primary magnetization. They yield a paleomagnetic pole located at lat. 41° N, long. 237° E ($n = 6$, $k = 186°$, $\alpha_{95} = 5°$). This pole position implies the existence of a large upper Jurassic–lower Cretaceous swing in the Iberian polar-wander path. The comparison with other polar paths (North America, Africa) supports the hypothesis of an initial lower Cretaceous rotation of Iberia prior to the main mid-Cretaceous shift. The remainder of the components represent remagnetizations. The major phase (groups 2 and 3) extended from the upper Cretaceous to the Eocene at least, a period closely correlated with the period of convergence between the European and Iberian plates. The directions of group 3 are characterized by low inclination. This is a common behaviour for Eocene directions in the Mediterranean area. Previous data obtained from the Wealden redbeds largely agrees with the directions of one or other of the groups 2 and 3. As there is strong evidence of an acquisition subsequent to the main mid-Cretaceous movement of Iberia, the geodynamical inferences of the paleomagnetic data have to be re-examined. The last type of components (group 4) was built up during recent remagnetization and is of little interest.

Introduction

Recent developments in the kinematics of the Iberian plate have stressed the importance of the lower Cretaceous in its evolution. Therefore, it is of major importance to get paleomagnetic data for this period. The continental deposits known as the Wealden facies, outcropping in the Cantabrian Chain and in the Sierra de los Cameros at the northwestern end of the Iberian chain, offer a good opportunity for a paleomagnetic study. Previous results obtained by Stauffer and Tarling (1971) from one site and Van den Berg (1979) from two sites were merely interpreted as antedating the rotation of the Iberian Peninsula, contemporaneous with the opening of the Bay of Biscay. This assumption supposes (1) that the measured magnetic directions are of primary origin and (2) that the European paleomagnetic poles of similar age are fairly well defined. Unfortunately, the second condition is far from being realized as can be observed, for example, in a recent compilation made by Westphal et al. (1985). We will be dealing later on with the first condition.

In order to increase the reliability of the lower Cretaceous Iberian paleomagnetic pole, we have carried out a more extensive study of the Wealden series, mainly in the Sierra de los Cameros, where the sedimentary basin is best developped and the formations well exposed. As regards the intensity

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of magnetization, redbeds are suitable material for paleomagnetic studies, but remagnetization processes often occur, so that an extensive sampling is highly desirable. This proved to be a safe procedure in this study, which revealed that the magnetic history of the Wealden series is much more intricate than expected.

Geology

The continental sediments deposited in the lower Cretaceous in northern Spain followed the Jurassic marine sedimentation. They are indicative of a distensive phase beginning in the Pyrenean and Cantabrian area in the upper Jurassic. The sediments accumulated in two basins: the Cantabrian basin to the north and the Soria basin or Sierra de los Cameros to the south. Salomon (1982) distinguishes three sedimentary megacycles, the first one being represented by the Wealden series.

Stratigraphy and ages

Salomon (1982) gives a detailed description of megacycle 1 in both areas. He uses two different nomenclatures to describe the lithologic subdivision of the deposits. This apparent inconsistency merely emphasizes the fact that the basins were never connected and, therefore, that it is impossible to correlate the sedimentary sequences identified in each of them. In the Cantabric chain (Fig. 1), Salomon divides megacycle 1 into three units, the third unit being itself divided into three terms 3A, 3B and 3C. The lowermost term is generally represented by conglomerative beds. These pass into finer-grained clastics such as sandstones, siltstones, argillites and marls, variegated or red to red–violet colored. Lacustrine, fossiliferous limestones are present at different stratigraphic levels, mainly in the area of Arroyal. In the area of Ramalès, an important marine episode took place at the top of term 3A. The formations covering the Wealden series and attributed to megacycle 2 begin either with limestones, as in the area of Ramalès where they have an assigned Aptian age, or with sandstones, as in the areas of Vega de Pas or Arroyal.

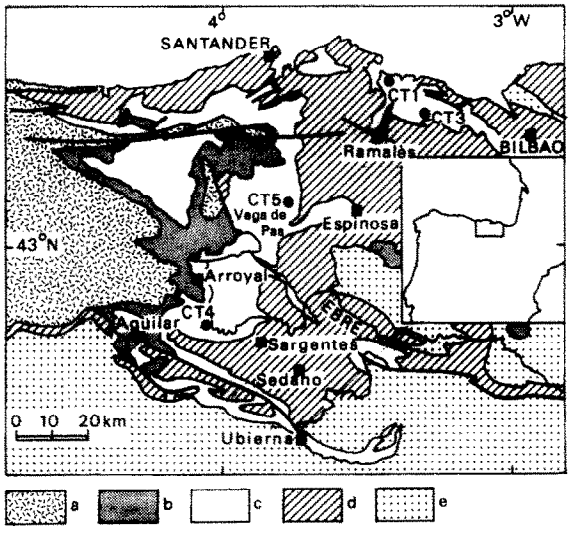


Fig. 1. Geological map of the Cantabric Chain and locations of the sampling sites (after Salomon, 1982, simplified). Key: *a* = Paleozoic and Permo–Triassic; *b* = Keuper and Jurassic; *c* = continental deposits (upper Jurassic–upper Albian); *d* = upper Cretaceous; *e* = Tertiary.

In the Sierra de los Cameros, megacycle 1 is very well developed, reaching an overall thickness of 7000 m. Salomon distinguishes four units, named A, B, C and D (Figs. 2 and 3). They consist of conglomerates, sandstones, siltstones, argillites, marls, and lacustrine fossiliferous limestones. The fine-grained beds are often red or violet–red colored. One observes frequent lateral heterogeneity of facies, especially in terms A and D, and a great variation in thickness, caused by active synsedimentary faults oriented N60 or N135.

The dating of the different terms constituting megacycle 1 is based on the ostracods fauna and Charophytes. But there is no general agreement about the suggested ages (see Salomon, 1982, for a review of the various existing scales).

A notable case of controversy occurs at Vega de Pas (Fig. 1) regarding term 3C, which was sampled by Van den Berg (1979) and ourselves. Salomon (1982) has suggested a Valanginian age, whereas Pujalte (1977) has estimated that it is Hauterivian. As for Van den Berg (1979), he assumed that the sampled levels belong to the Barremian–Aptian. His assessment is based on the determination of orbitolinas present in limestone

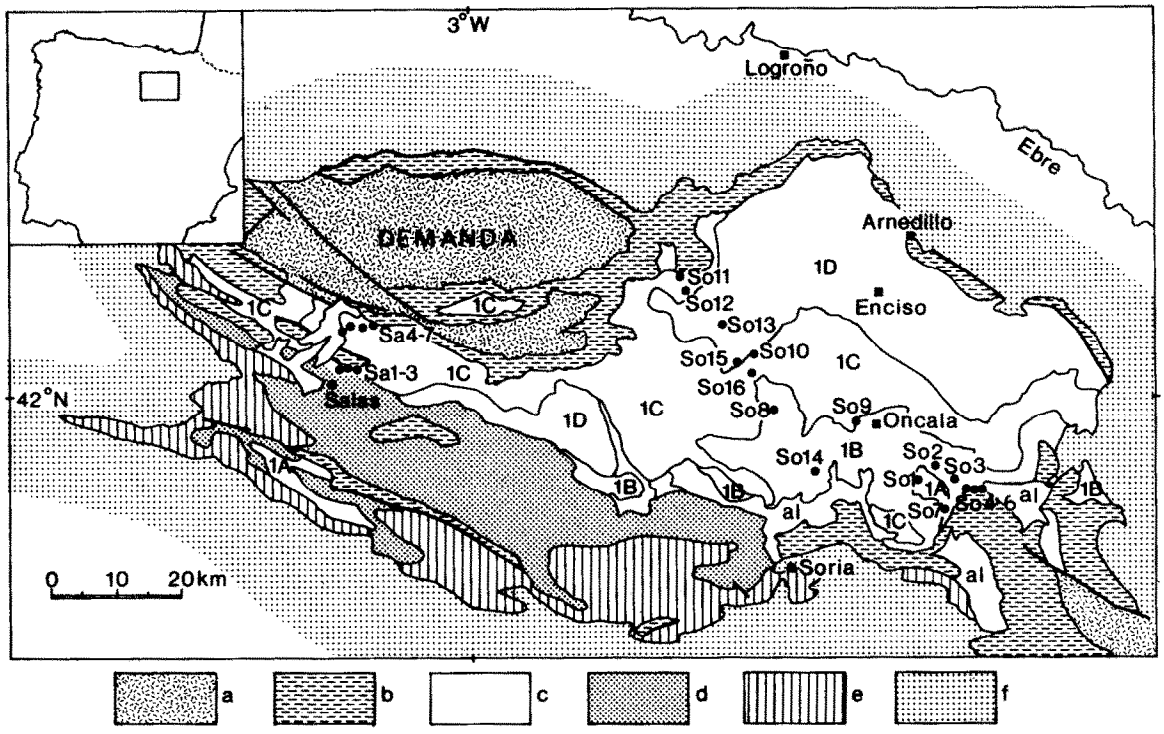


Fig. 2. Geological map of the Sierra de los Cameros and locations of the sampling sites (after Salomon, 1982, simplified). Key: *a* = Paleozoic and Permo–Triassic; *b* = Upper Triassic and Jurassic; *c* = Wealden series (1A, 1B, 1C and 1D); *d* = Barremian–Albian continental deposits; *e* = Utrillas and marine Upper Cretaceous; *f* = Tertiary; *al*—alluvium.

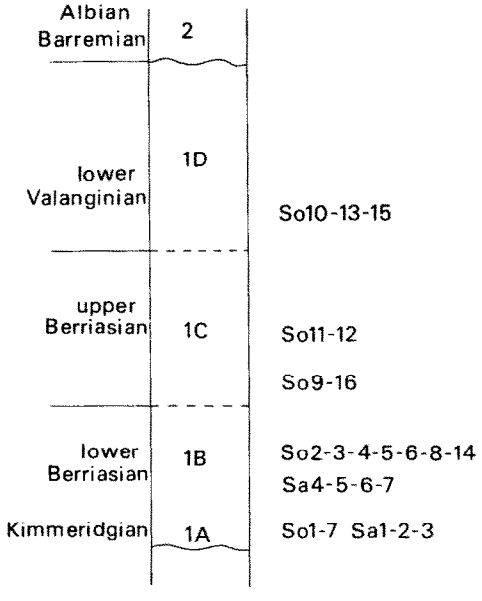


Fig. 3. Distribution of the sampling sites from the Sierra de los Cameros on the stratigraphic column of Salomon (1982).

intercalations at the top. But in the description of Salomon (1982), the first limestones to appear lie upon the coarse sandstones representing the bottom of megacycle 2 (Urgonian complex). Thus, the estimate adopted by Van den Berg (1979) seems too young.

To keep a coherent scale, we will adopt the dating proposed by Salomon (1982) throughout this paper. The distribution of the sampled levels from the Sierra de los Cameros on this author's stratigraphic scale is shown in Fig. 3. The sampling sites from the Cantabric Chain (Fig. 1) belong to term 3B (site CT4) or 3C (sites CT1, CT3 and CT5). The assigned ages are Valanginian (site CT5) or Valanginian–Hauterivian (sites CT1, CT3 and CT4).

Tectonics

Cantabric Chain

In the area of Vega de Pas, the strata dip gently towards the southeast. It is rather difficult to

assess the age of this moderate deformation. It is probably mainly due to Tertiary tectonics. The deformation is also apparently very moderate in the area of Ramalès. The sampled beds of sites CT1 and CT3 dip slightly towards the west or south. This area belongs to the Bilbao anticlinorium, although the Tertiary deformation is greatly reduced here. Nevertheless, decollements of the Mesozoic cover, well known in the whole Basque arc, also affect the region of Ramalès. Therefore, one must be careful when interpreting the paleomagnetic data from this zone. Synsedimentary deformation is obvious in the area of Arroyal, mainly from the deposition of term 3A onwards, and remains active during the sedimentation of megacycles 2 and 3. In addition, deformation dating from the lower Tertiary took place on the southern border of the region, especially south of the Ubierna fault zone (Fig. 1). North of this fault, the influence of Tertiary tectonics is clearly reduced so that the deformation is essentially synsedimentary in the locality studied.

Sierra de los Cameros

Two major phases of deformation occurred in this area: a compressional phase, dating from the Oligo-Miocene, and a phase coeval with the development of the sedimentary basin. The most recent deformation is best observed in the western part of the basin, and on the northern and northeastern edges. It is linked with the strike-slip fault active in the Iberic chain during the Oligo-Miocene (Canerot et al., 1982). The uplift of the Paleozoic Demanda massif is also connected with this Tertiary movement. In particular, one can attribute an Oligo-Miocene age to the tilt of the sampled beds north of Salas de los Infantes.

The synsedimentary deformation clearly affected the eastern part of the basin and can be observed occurring at different scales. It gave rise to a moderate folding whose axes are oriented N135, and are cut by faults and flexures oriented N60. The folding is slightly tighter at the N60-striking borders. On the southeastern edge, the folding is emphasized by schistosity (Guiraud, 1983). This author pointed out that the successive beds were deposited horizontally and that the dipping occurred everywhere after the sedimentation. The word synsedimentary must be understood in a rather broad sense because the movements continued during megacycle 2 (Barremian–Albian). They diminished progressively with the third megacycle, which is characterized by uniform sedimentation covering the whole basin and spreading far beyond the basin.

A low-grade metamorphism affected the formation situated east of the line Arnedillo–Oncala, described by Guiraud (1983) as syn- to post-schistous. It gave rise to a recrystallization of sandstones in quartzites and to hardened levels in siltstones. In the same zone, one can observe occurrences of cubic pyrite. Metamorphism and mineralization are present in terms ranging from 1B to the lower part of 1D, but apparently they do not reach the upper part of 1D, the Olivan group (Fig. 3).

Sampling and analytical techniques

The sampling of the Wealden series was carried out during two successive field seasons (1979 and 1980) and included 27 sites (4 in the Cantabric chain and the remaining ones in the Sierra de los Cameros). In addition, two formations belonging to the basal levels of megacycle 2 (Aptian–Albian in age) were sampled in the Cantabric Chain (in the vicinity of sites CT1 and CT5, respectively), consisting of grey sandstones. But these display a very weak and scattered magnetization, so that these sites will be discounted from further consideration. The same remark holds true for site SO10 (Fig. 2), which consists of yellow sandstones belonging to term 1D (Fig. 3).

In all other cases, the sampled beds consist of red to purplish–red or variegated sandstones. As far as possible, the samples were evenly distributed over the outcrop and over a stratigraphic thickness as widely extended as possible. The height varied from 0.5 m up to about 30 m. The sites were hand sampled or drilled. The samples were oriented with magnetic and sun compasses. Two or three standard core samples, 2.5 cm in diameter and 2.5 cm in length, were obtained from each sample.

The NRM of the specimen was measured with a "Digico" fluxgate magnetometer. At least one

specimen per sample was submitted to progressive thermal demagnetization consisting of at least eight steps up to 680 °C. The susceptibility was measured initially and after each step of treatment with a commercial ac bridge (Digico) in order to detect the mineralogy changes induced by heat.

In addition, specimens from samples showing a complicated thermal demagnetization curve were selected for chemical leaching. The procedure used was as follows: a hole 6 mm in diameter was drilled along the axis of the cores; the cores were then submitted to a thermal demagnetization at 180 °C in order to eliminate the viscous component. The chemical leaching was performed with 6 *N* hydrochloric acid, whose concentration was increased, as the treatment progressed, up to 12 *N*. During leaching, the specimens were placed in a magnetic shield reducing the vertical component of the ambiant field to 70 nT. This value was low enough to prevent the acquisition of a viscous remanant magnetization (VRM). The cumulative immersion time reached 1000 h in some cases, but, generally, no significant change occurred after 600–700 h.

Analysis of NRM; classification of magnetic components

As a general rule, thermal treatments were more efficient than chemical ones for separating the magnetic components of the NRM. As pointed out by some authors using the acid leaching technique (e.g., Henry, 1979; Turner, 1981), the difficulty inherent in this method is to obtain uniform leaching throughout the sample in order to remove progressively coarser grains. The difficulty is increased still further if authigenic overgrowths protect pigmentary grains from acid contact. Whatever the reason, we observed a consistently greater overlap of the dissolution rates of the different components than of the blocking-temperature spectra.

Figure 4 illustrates this observation. The thermal demagnetization (Fig. 4a) allows us to define 3 components, indicated by triangles on the stereographic plot. By chemical leaching (Fig. 4b), two components are fairly well revealed, the softest being calculated by vector substraction between

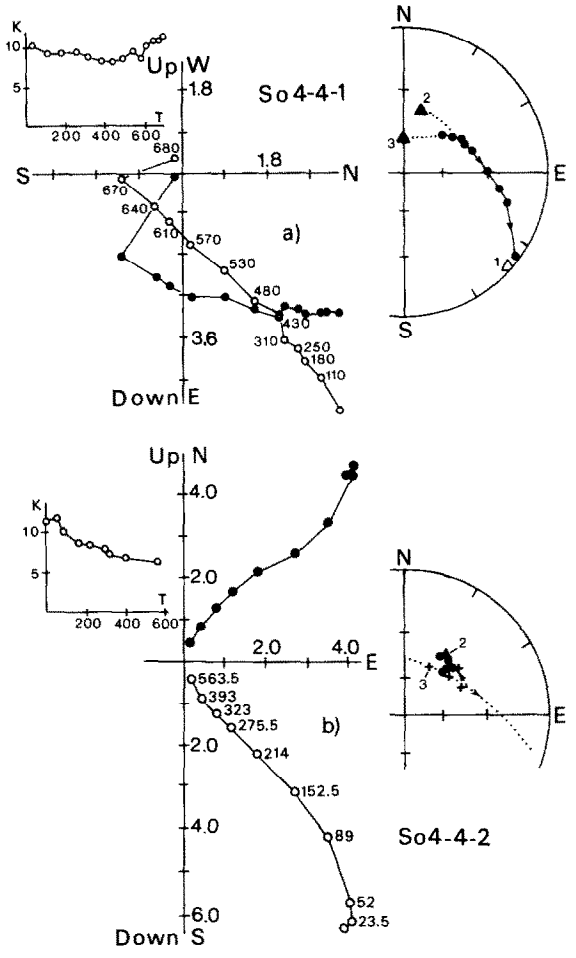


Fig. 4. Comparison of (a) thermal and (b) chemical demagnetizations. On the orthogonal diagrams, solid circles are projections onto the horizontal plane and open circles are projections onto the vertical plane. Units along the axes are in 10^{-3} A m^{-1}. On the Wulff-net plot of the demagnetization paths, solid circles are projections onto the lower hemisphere. Solid triangles represent the directions of the components of the magnetization (see text). Crosses are the directions of the difference vectors in the range 52–214 h. K = susceptibility per unit volume in 10^{-5} S.I. h = cumulative hours of leaching, T = temperature.

the points 52 h and 89 h (the directions of the difference vectors are indicated by crosses on the stereogram—the softest component corresponds to cross 3), the hardest by a least-squares line using the three last points of the curve (solid triangle labeled 2). They are similar to components 3 and 2, respectively, of the thermal cleaning. Both the Zijderveld diagram and the stereographic plot show the existence of a third compo-

nent. It can be loosely characterized by the great circle passing through the subtraction vectors calculated between points 52 h and 214 h. The great circle approaches the direction of component 1 from Fig. 4a. Thus, both demagnetization methods reveal the same composition of natural remanant magnetization (NRM), but in the chemical method, there is a considerable overlap between the dissolution rates of components 1 and 2, and 1 and 3 (thermally defined). Furthermore, component 1, possessing the highest blocking temperatures (Fig. 5) is carried by minerals easily removed by acid leaching. Figure 4b shows also that the susceptibility decreases more slowly than the NRM. In redbeds with magnetic components carried by pigment and specularite, or by pigment alone, the lost of leachable iron follows the same variation (Turner, 1981).

The magnetic components, mainly determined by thermal stepwise demagnetization, are classified into 4 groups, which are defined as follows:

(1) Group 1 contains normal and reversed directions yielding a mean declination/inclination of 319°/18°, after unfolding.

(2) Group 2 consists of normal polarities with a mean $D = 351°$, $I = 47°$, after tectonic correction.

(3) Group 3 has normal polarities although some samples reveal the presence of a high temperature reversed component; the directions cluster around $D = 359°$, $I = 31°$, without unfolding.

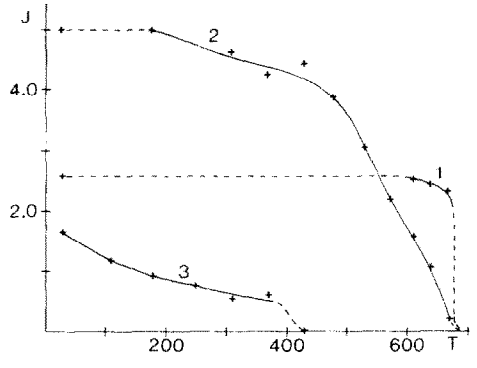


Fig. 5. Thermal decay curves of the different magnetization components of the specimen from Fig. 4a. *1*, *2* and *3* refer to the three components identified on that figure. *J* is the intensity of magnetization in 10^{-3} A m^{-1} and *T* is the temperature in °C.

(4) Group 4 comprises merely the recently acquired components.

These 4 types of magnetic components are tentatively classified in decreasing ages from 1 to 4. The reasons which support this ordering and the assessment of ages of magnetization will be discussed in the next section. Components of type 2 or 3 may be present as single components, giving rise to a simple demagnetization curve. Components of type 1 are always superimposed on a component of type 2 or 3, and always possess the highest blocking temperatures (above 600°–650° C).

Discussion of the results

Group 1 (Table 1 and Fig. 6)

Because of the presence of both polarities, the main question is whether the magnetization was acquired before or after the long normal period of the Cretaceous (Larson and Hilde, 1975). The fold test (Graham, 1949) may first give evidence. Bearing in mind that two main tectonic episodes occurred, three assumptions have to be examined: (a) the magnetization is everywhere post-tectonic, (b) it is posterior to the synsedimentary deformation but anterior to the Tertiary folding, (c) it is penecontemporaneous with the sedimentation. The condition required by the statistical test of McFadden and Jones (1981) not being fulfilled here, we applied the older test tabulated by McElhinny (1964) and based on Watson's statistics (1956). It shows that the fold test is not significant between assumptions (b) and (c), but it becomes positive, at the 95% level, between assumptions (a) and (c) if we discard the directions of sites SA7 and CT1 (Fig. 6). The results of site CT1 will be discussed later. The rejection test of McFadden (1982) makes direction SA7 appear as an outlier (after tectonic correction only). But apart from this statistical consideration, there is no obvious reason to reject it. In short, the fold test brings evidence against assumption (a) but does not allow us to choose between (b) and (c).

The most reliable argument is obtained from the comparison of the paleomagnetic poles yielded by the group with known apparent polar-wander

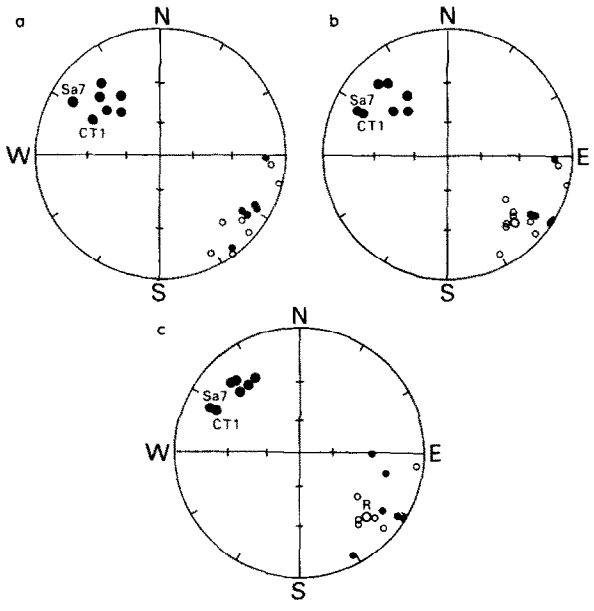


Fig. 6. Stereographic plots of the directions of group 1. (a) before tectonic correction; (b) before synsedimentary tectonic correction and after lower Tertiary tectonic correction; (c) after tectonic correction everywhere. Small circles are individual directions, larger circles are site mean directions. Solid circles are projections onto the lower hemisphere, open circles are projections onto the upper hemisphere. *R* — reverse mean directions. The fold test is significant between (a) and (c) when the direction Sa7 is discarded. It is not significant between (b) and (c).

paths (APW). This comparison needs the reassembly of the corresponding plates according to given kinematics. We used the parameters calculated by Olivet et al. (1984), and moved the plates in a frame fixed relative to the present position of Iberia. The best example is afforded by the APW of North America (Harrison and Lindh, 1982).

To transfer this curve to Iberia we combined the kinematic parameters of the relative movements of North America/Europe and Europe/Iberia, respectively. The movement of Iberia relative to the European plate is still a matter of debate (Souquet and Debroas, 1980; Boillot, 1984; Masson and Miles, 1984; Olivet et al., 1984). Olivet et al. (1984) suggested the occurrence of a counterclockwise rotation of Iberia prior to its major movement in the Cretaceous, coeval with the opening of the Bay of Biscay. But these authors did not suggest kinematic parameters for this shift, so we computed a tentative pole of rotation: lat. 46.9° N, long. −15.9° E, angle = 15°.

The timing of the initial rotation is largely unknown. Other authors have postulated movement prior to the opening of the Bay of Biscay (Malod, 1982; Boillot, 1984; Masson and Miles, 1984), between the Valanginian and the Albian, but the inferred directions of motion are quite different from each other. Without entering into detailed discussion of the paleomagnetic implications of the different models, we will examine the consequences of the initial rotation for the location of the lower Cretaceous paleomagnetic pole.

Figure 7 displays the transfered APW of North America according to the kinematics of Olivet et al. (1984). The solid portion between the upper Jurassic and the Middle Cretaceous is drawn on the assumption that the initial rotation is posterior to the lower Cretaceous, the dotted portion on the alternative assumption. The first hypothesis gives the APW curve a marked "hairpin" shape, reaching its peak in the lower Cretaceous. The lower Jurassic and upper Cretaceous poles located at the beginning and end of the hairpin are close to one another (but the duration of the anomalous behaviour is, at least partly, artificially extended by the smoothing method used by Harrison and Lindh, 1982). Under the alternative assumption (dotted line), the hairpin is much less pronounced. As for the comparison of the paleomagnetic pole yielded by group 1 (labeled 1 in Fig. 7), its far-sided location is fairly similar to the position of the

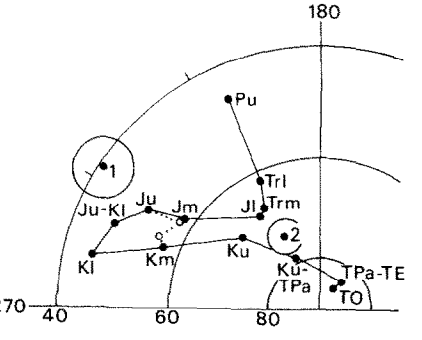


Fig. 7. Transferred APW of North America (after Harrisson and Lindt, 1982) in a frame fixed to Iberia according to the kinematics of Olivet et al. (1984). See text for the difference between the solid portion and the dotted portion of the curve. *1* is the mean pole for group 1; *2* is the mean pole for group 2.

lower Cretaceous pole of North America. We suggest, therefore, (1) to attribute an upper Jurassic–lower Cretaceous age to the directions of group 1, and (2) to place the initial rotation after the build-up of magnetization. Hence, the components of group 1 represent the primary magnetization of the Wealden series. Furthermore, we argue that it everywhere predates the synsedimentary deformation. Bearing in mind that the synsedimentary deformation extended beyond megacycle 1, however, this assessment does not imply any close correlation between sedimentation processes and magnetization. Examination of Table 1 and Fig. 3 shows that the primary magnetization is carried by levels belonging to the Kimmeridgian and the lower Berriasian. Thus, the postulated hairpin of the Iberian APW is located in a very restricted period, consistent with a supposed Valanginian–Albian age of the initial motion of Iberia.

The comparison with the APW of North America affords the most convincing evidence for assessing the date of the primary magnetization because this APW is well documented. The postulated hairpin appearing on the Iberian curve in the upper Jurassic–lower Cretaceous is supported by further comparisons demonstrating the worldwide extent of this feature. Whatever compilation of African data is used (e.g., Van den Berg, 1979; Besse et al., 1984; Westphal et al., 1985), the APW of Africa exhibits a similar hairpin in the lower Cretaceous. Another example is given by Marton and Marton (1983) for the Transdanubian Central Mountains. As a final example, a compilation of the numerous data available for Umbria (Lowrie and Alvarez, 1977; Alvarez and Lowrie, 1978; Van den Berg et al., 1978; Van den Berg, 1979; Lowrie and Channell, 1983; Channell et al., 1984) gives rise to the same comments: the paleomagnetic poles of the middle Jurassic and upper Cretaceous are close to one another, as has already been pointed out by Channell et al. (1984), whereas the poles of the upper Jurassic are shifted. This feature can be interpreted as the occurrence of a "hairpin". This proposal is further supported by the work of Cirilli et al. (1984).

TABLE 1

Site mean directions and paleomagnetic poles of group 1 *

Site	n/n_0	Site mean directions				Statistics			Paleomagnetic poles [d]			
		without tect. cor.		with tect. cor.								
		D_m (deg.)	I_m (deg.)	D_m (deg.)	I_m (deg.)	N	k	α_{95} (deg.)	lat. (°N)	long. (°E)	dp (deg.)	dm (deg.)
CT1	13/13	297	28	296	16	4	33	16	24	254	8	16
SA4	9/8	314	23	315	14	7	38	10	37	238	5	10
SA7	7/6	301	12	296	13	6	27	13	24	254	7	13
SO1	9/9	320	17	318	15	3	134	11	39	237	5	11
SO3	7/7	309	32	322	21	4	61	12	45	236	6	12
SO9	10/9	318	41	316	22	9	34	9	41	242	5	9
SO16	7/7	326	31	329	21	5	128	7	48	228	4	7
SA1, SA2, SA5 SA6,SO4, SO5 [a] SO6, SO7	$n = 13$ [b]	136	−5	133	−17	5	64 [c]	9 [c]	37	242	5	10
Mean direction and poles (CT1 and SA7 excluded)	without tect. cor.:	$D_m = 317°$		$I_m = 25°$								
		$N = 6$		$k = 36$		$\alpha_{95} = 11°$						
	with tect. cor.:	$D_m = 319°$		$I_m = 18°$					N lat. = 41°, E long. = 237°			
		$N = 6$		$k = 157$		$\alpha_{95} = 5°$			$k = 186$ $\alpha_{95} = 5°$			

* Notes: n/n_0 = number of specimens/number of samples; D_m, I_m = mean declinations and inclinations; N = number of directions used in the Fisher statistics; k, α_{95} = parameters of Fisher statistics; dp, dm = semiangles of the ellipse of confidence; [a] Sites where reversed polarities are present, [b] Total number of reversed directions. [c] Statistics calculated after tectonic corrections. [d] Poles calculated after tectonic corrections.

We must nevertheless point out an important exception: so far, no anomalous directions have been mentioned in localities belonging to the European plate. This phenomenon is further supported by the magnetostratigraphic study carried out by Galbrun (1985) on the Berriasian stratotype section at Berrias (France). The relative motions of the different plates may partly explain this discrepancy, but a significant divergence still persists between the lower Cretaceous poles of Europe on the one hand and those of Africa, North America and South America on the other hand, after replacing the plates in their respective locations (Westphal et al., 1986).

Group 2 (Table 2 and Fig. 8)

Only normal polarities are present in this group. The statistical test of McElhinny (1964) is performed separately for the directions of sites CT5 and SA(n) (Fig. 8a) and for the directions of sites SO(n) (Fig. 8b) because the foldings have differ-

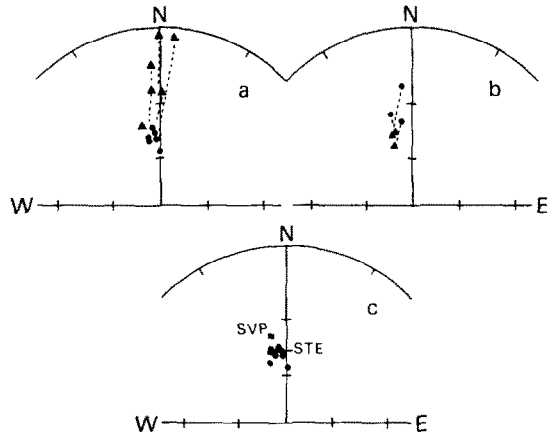


Fig. 8. Stereographic plot of the site mean directions of group 2: (a) sites with lower Tertiary tectonics; (b) sites with synsedimentary tectonics; (c) characteristic directions of group 2. Solid triangles are the direction before tectonic correction, solid circles are the directions after tectonic correction (all directions are in the lower hemisphere). SVP and STE are the site mean directions obtained by Van den Berg (1979).

TABLE 2

Site mean directions and paleomagnetic poles of group 2

Site	n/n_0	Site mean directions				Statistics			Paleomagnetic poles [b]			
		without tect. cor.		with tect. cor.								
		D_m (deg.)	I_m (deg.)	D_m (deg.)	I_m (deg.)	N	k	α_{95} (deg.)	lat. (°N)	long. (°E)	dp (deg.)	dm (deg.)
CT5	5/5	346	41	350	48	5	82	8	74	210	7	11
SA1	10/9	5	3	356	48	10	52	7	77	191	6	9
SA2	10/9	356	14	354	43	7	252	4	72	195	3	5
SA3	7/6	0	3	0	55	7	176	4	83	174	4	6
SA6	7/6	356	24	350	48	4	201	6	75	213	5	8
SA7	7/6	1	26	354	45	7	92	6	74	196	5	8
SO3	7/7	345	45	355	22	7	219	4	70	221	3	5
SO13	5/5	343	51	347	34	5	542	3	74	236	3	4
SO14	5/5	347	44	353	39	5	633	3	71	215	2	4
SVP [a]	–	–	–	349	38	–	–	1	66	202	1	1
STE [a]	–	–	–	352	45	–	–	4	–	–	–	–
Mean directions and poles (STE excluded)	without tect. cor.:		$D_m = 354°$		$I_m = 28°$							
			$N = 9$		$k = 17$		$\alpha_{95} = 12°$					
	with tect. cor.: (only applied on sites SA, CT5 and SVP)		$D_m = 351°$		$I_m = 47°$				N lat. = 74°, E long. = 207°			
			$N = 10$		$k = 196$		$\alpha_{95} = 3°$		$k = 173$ $\alpha_{95} = 4°$			

[a] Data of Van den Berg (1979).

[b] Poles calculated with tectonic corrections for sites SA and SVP, without tectonic corrections for sites SO.

Other symbols as in Table 1.

ent ages in the corresponding areas. In the first case, the test is positive at the 99% level; in the second case, it is negative at the 90% level owing to the small number of directions. So the fold test shows clearly that the magnetization predates the Oligo–Miocene phase of deformation, but postdates the synsedimentary deformation. The comparison with the APW of North America, Africa and Europe, returned to their predrift locations, shows that the paleomagnetic pole of this group (labeled 2 on Fig. 7) is near the upper Cretaceous poles. Its position is also close to the pole yielded by the Lisbon volcanics (Van der Voo and Zijderveld, 1971), whose K–Ar age is 73 Ma according to Ferreira and Macedo (1979).

A last argument can be put forward based on the kinematics of Iberia: we may suppose that the magnetization was built up before the main middle Cretaceous movement. The relocation of the pole prior to this motion (relative to Europe for example) gives a position near the lower Jurassic pole of Europe. This consequence shows the inconsistency of the assumption.

Therefore, we attribute to the magnetization of group 2 an upper Cretaceous age. The component represents a first phase of remagnetization superimposed upon the original magnetization. The stages of remagnetization are variable: a partial remagnetization is present everywhere, but at some places this phase has completely replaced the primary one. Figure 8c shows that the site mean directions labeled SVP and STE, representing the data obtained by Van den Berg (1979) at Vega de Pas (our site CT5) and Tera (some 4 km east of our site SO8), respectively, are very similar to the directions of group 2. That author claimed that the direction SVP just predated the major motion of Iberia (contemporaneous with the opening of the Bay of Biscay) and he attributed it to a Barremian–Aptian age. Our data do not support this conclusion but lead us to the reinterpretation suggested above.

Group 3 (Table 3 and Fig. 9)

Similarly to group 2, we have to consider two sets of data in connection with the age of folding. Unfortunately, there are only two sites (SA4, SA5)

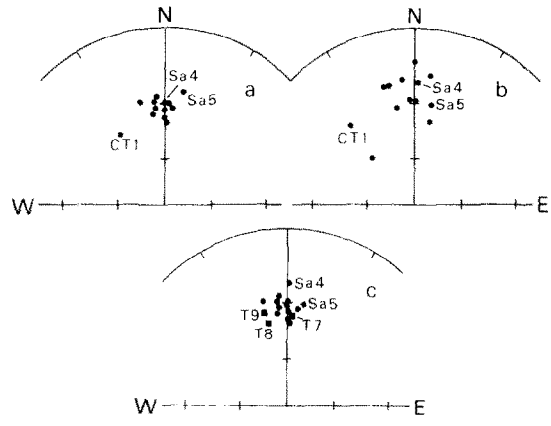


Fig. 9. Stereographic plot of the site mean directions of group 3: (a) before tectonic correction; (b) after tectonic correction; (c) characteristic directions of group 3. All directions are in the lower hemisphere. T7, T8 and T9 are the site mean directions obtained by Stauffer and Tarling (1971).

in the area affected by the Tertiary deformation. In addition, the tilt corrections are small so that the fold test is inconclusive in this case. On the other hand, the statistical test of McElhinny (1964) is negative at the 99% level for the sites SO(n). But this result merely demonstrates that magnetization postdates synsedimentary tectonics. The comparison with paleomagnetic poles of North America, Africa and Europe, prior or posterior to the mid-Cretaceous motion of Iberia, shows clearly that the magnetization of group 3 postdates this major phase of evolution. Actually, the distinction between directions of group 2 and 3 is somewhat arbitrary. It seems to us that they are the result of a long period of remagnetization, the beginning of which is represented by group 2. The distinction introduced here is based mainly on the inclination, those of group 3 being definitely lower. This characteristic can be connected with observations made by Westphal et al. (1986) throughout the Mediterranean realm: these authors point out that low inclinations occur in many Eocene-aged localities.

The occurrence of low inclinations in the lower Tertiary may also be connected to long-term nondipole fields analyzed with spherical harmonic analysis of data sets extending back 130 Ma (Coupland and Van der Voo, 1980) or 200 Ma (Livermore et al., 1984; Lee et al., in prep.). The first and the third studies can be more readily

TABLE 3

Site mean directions and paleomagnetic poles of group 3

Site	n/n_0	Site mean directions				Statistics			Paleomagnetic poles [b]			
		without tect. cor.		with tect. cor.								
		D_m (deg.)	I_m (deg.)	D_m (deg.)	I_m (deg.)	N	k	α_{95} (deg.)	lat. (°N)	long. (°E)	dp (deg.)	dm (deg.)
CT1	13/13	326	39	321	30	8	172	4	47	240	3	5
SA4	9/8	3	30	1	21	10	251	3	59	175	2	3
SA5	9/7	10	24	9	30	8	189	4	63	158	2	4
SO1	9/9	354	29	349	31	12	150	3	63	190	2	4
SO2	8/7	1	40	11	39	8	122	5	70	174	3	6
SO5	11/9	6	32	6	17	11	88	5	65	165	3	5
SO6	10/9	0	32	359	12	12	67	5	66	178	3	6
SO7	9/9	356	27	347	21	12	57	6	62	187	3	6
SO8	9/9	0	29	0	29	11	130	4	63	178	2	4
SO9	10/9	353	35	346	21	2	–	–	66	194	–	–
SO12	10/9	347	28	318	50	4	67	11	60	204	7	12
SO15	5/5	0	37	358	29	5	279	4	69	177	3	5
SO16	7/7	354	31	354	20	7	330	3	65	190	2	4
T7 [a]	–	–	–	2	36	9	127	5	–	–	–	–
T8 [a]	–	–	–	347	39	8	116	5	–	–	–	–
T9 [a]	–	–	–	346	34	6	542	3	–	–	–	–
Mean directions and poles (CT1, T7, T8 and T9 excluded)	without tect. cor.:			$D_m = 359°$	$I_m = 31°$							
				$N = 12$	$k = 136$			$\alpha_{95} = 4°$				
	with tect. cor.: (sites SA only)			$D_m = 358°$	$I_m = 31°$				N lat. = 65°, E long. = 181°			
				$N = 12$	$k = 128$			$\alpha_{95} = 4°$	$k = 155$	$\alpha_{95} = 4°$		

[a] Data of Stauffer and Tarling (1971); [b] Poles calculated with tectonic corrections for sites CT1 and SA, without tectonic corrections for sites SO.

Other symbols as in Table 1.

compared, as both attempt to evaluate the variations of $G_2°$ and $G_3°$ with time. Although there are noticeable differences in the results, both analyses yield relatively high values of $G_2°$ and $G_3°$ at about 40 Ma. So a postulated Eocene date for the magnetization of group 3 is consistent with the behavior of the Earth's field determined on a world-wide scale.

The directions labelled T7, T8 and T9 on Fig. 9c were obtained by Stauffer and Tarling (1971) in the neighbourhood of Tera. Their original paper did not mention whether the data were given after unfolding. But as the corrections are negligible in this area, this uncertainty does not affect our interpretation, which classifies those directions in group 3.

Group 4 (Table 4 and Fig. 10)

The magnetic components pertaining to this group are characterized by blocking temperatures lower than 300 °C, apart from the site CT1 where high temperatures are needed to remove them. Figure 10 shows that the directions definitely postdate the folding and that most of them cluster around the present-day field. Some distinction could be made for the directions SO4, SO11 and SO12, which give paleomagnetic poles close to the Eocene or Oligo-Miocene poles of Europe or Africa. But we will not argue about such details. This could serve as further evidence for a long-lasting remagnetization phase extending from the upper Cretaceous to the Oligo-Miocene.

Sites CT1 and CT3

The particular location of these sites (Fig. 1) calls for special consideration. In addition to recent components, Tables 1 and 3 and Fig. 6 and 9 show that components of groups 1 and 3 are present at the site CT1. The existence of primary

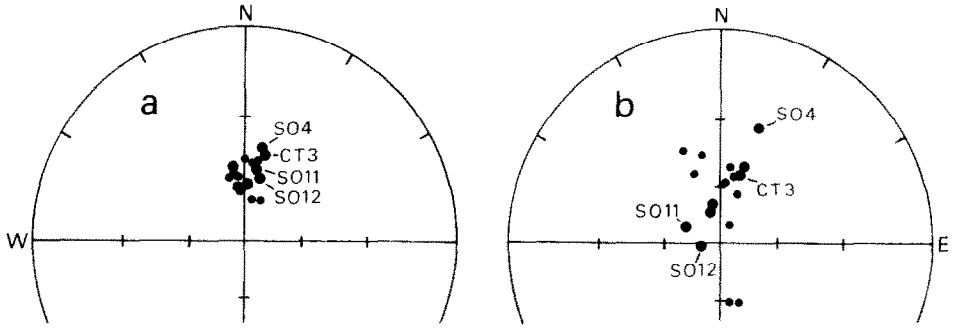


Fig. 10. Stereographic plot of the directions of group 4: (a) before tectonic correction; (b) after tectonic correction. All directions are in the lower hemisphere. Small circles are individual directions, larger circles are site mean directions.

magnetization is questionable because, contrary to all other cases, these components have low blocking temperatures, and the sampled beds belong to the youngest formation of the Wealden series, with a postulated Valanginian–Hauterivian age. But due to the difficulties inherent in the stratigraphic correlation in this area, this is not a conclusive argument against our interpretation. In any case, however, the declinations of site CT1 show a significant departure from the corresponding group means, amounting to about 30°–40°. This difference could be interpreted as a counter-clockwise rotation of the area, but this is not supported by the data from site CT3, which does not differ significantly from the mean (Table 4 and Fig. 10). Of course, we could suppose that in this latter area the magnetization postdates the hypothetical rotation, but there is little evidence for this. Let us recall that the tectonics of this area is rather complicated; in particular, there is no clear relationship between the sedimentary cover and the basement. So we must be very careful in inferring crustal movements from observations made in the cover.

TABLE 4

Site mean directions and paleomagnetic poles of group 4

Site	n/n_0	Site mean directions				Statistics			Paleomagnetic poles *			
		without tect. cor.		with tect. cor.								
		D_m (deg.)	I_m (deg.)	D_m (deg.)	I_m (deg.)	N	k	α_{95} (deg.)	lat. (°N)	long. (°E)	dp (deg.)	dm (deg.)
CT3(a)	13/10	14	46	15	54	6	110	6	70	137	5	8
CT3(b)		352	62	345	70	12	47	6	84	272	8	10
CT4	13/13	351	52	340	72	12	159	3	77	213	3	5
SO4(a)	6/5	11	43	18	31	5	267	5	71	146	3	6
SO4(b)		3	61	17	49	3	160	10	88	94	11	15
SO11	8/8	9	53	296	69	5	155	6	78	134	6	8
SO12	10/9	15	56	262	79	12	147	3	77	107	4	5
Mean directions and poles CT3(b), CT4, SO4(b), SO11 and SO12	without tect. cor.	$D_m = 2°$	$I_m = 57°$			$\alpha_{95} = 7°$	$k = 70$	$\alpha_{95} = 9°$	N lat. = 85°, E long. = 156°			
		$N = 5$	$k = 118$									
	with tect. cor.	$D_m = 340°$	$I_m = 72°$									
		$N = 5$	$k = 19$			$\alpha_{95} = 18°$						

* Poles calculated without tectonic corrections.

Other symbols as in Table 1.

Conclusions

This paleomagnetic study of the Wealden redbeds has shown itself to be much more complicated than expected from previous work. It gives a demonstrative example of the remagnetization capabilities of the redbeds. Figure 11 illustrates the distribution of the paleomagnetic poles yielded by the four groups of components and the inferred polar-wander path. Our analysis shows that the primary magnetization yielding the directions of group 1 is overprinted by a long-lasting remagnetization phase, extending from the upper Cretaceous to the Eocene, and perhaps to the Oligo-Miocene, in addition to the recent remagnetization. The original directions are characterized by rather surprising values of declination/inclination. The corresponding paleomagnetic pole shows up the existence of a large swing in the polar-wander path, quite similar to the behavior noted for other APW (North America, Africa, peri-Adriatic region) in the upper Jurassic–lower Cretaceous. The absence of this "hairpin" in Europe raises an interesting question about the origin of this feature. The discrepancy between European data and the data from other sources does not completely vanish after returning the plates to their upper Jurassic relative locations, so differential motions cannot entirely explain the phenomenon.

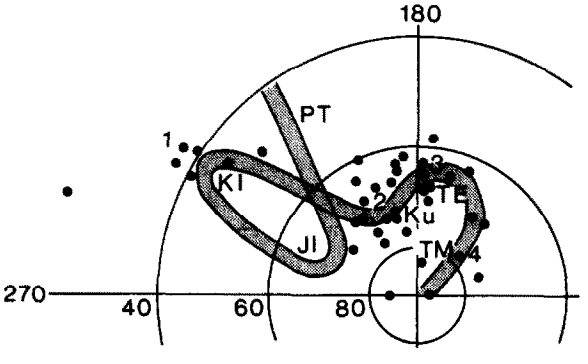


Fig. 11. Stereographic plot centered on the northern geographic pole, of the paleomagnetic poles yielded by the different classes of components. The shaded line is the suggested Iberian APW showing the period of acquisition of the different magnetic components carried by the Wealden series. Key: *PT* —Permo–Triassic; *Jl*—lower Jurassic; *Kl*—lower Cretaceous; *Ku*—upper Cretaceous; *TE*—Eocene; *TM*—Miocene.

The major remagnetization phase is represented by groups 2 and 3. Our conclusion is that these directions postdate the main sinistral motion of Iberia, coeval with the opening of the Bay of Biscay. We note that the postulated age of this remagnetization is, however, closely correlated with the period of convergence between the European and Iberian plates, inducing compression along their common boundary and leading to the Pyrenean orogeny. According to the kinematics of Olivet et al. (1984), the mid-Cretaceous movement of Iberia gave rise to the main rotation of the plate and, subsequently, to the principal effect on the position of the paleomagnetic poles. As we classify the data of Van den Berg (1979) and Stauffer and Tarling (1971) into one or other of the groups 2 and 3, our conclusions are quite at variance with theirs. The directions obtained by these authors, along with ours, can only offer clues to the movements of Iberia during the upper Cretaceous and more recent periods. However, according to the kinematics of Olivet et al. (1984), the total extent of rotation does not exceed 7.5°. To date, the accuracy of the APW of Europe and Iberia has not been good enough to allow an estimate of such a small value. This inability is increased by the occurrence of low inclinations in the lower Tertiary, as shown by group 3.

Figure 12 summarizes the relationship between the various magnetic directions found in the Wealden redbeds and the successive positions of the Iberian plate suggested by Olivet et al. (1984). The reference directions are deduced from the North American APW for the lower Cretaceous (Harrison and Lindh, 1982) and from the European APW for the upper Cretaceous and lower Tertiary (Westphal et al., 1986). This mixture of data is justified by the discrepancies between the reference APW's, discussed previously. Figure 12 shows that the primary directions alone may offer useful data for solving the kinematic problem. The comparison of the paleomagnetic pole with other APW support the assumption of an initial rotation subsequent to the build-up of magnetization. However, due to the existence of a large swing in the APW, this hypothesis requires further investigation of this crucial period in the evolution of the Iberian plate.

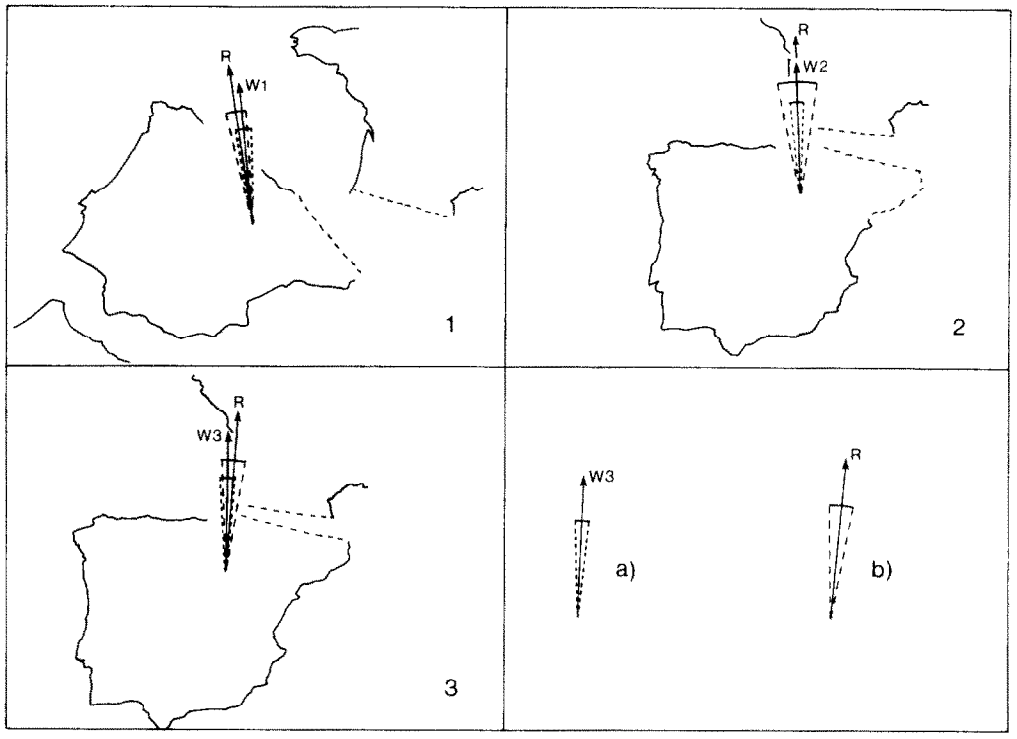


Fig. 12. Sketch map showing the relationship between the successive positions of the Iberian plate according to Olivet et al. (1984), modified in the upper Jurassic–lower Cretaceous (see text), and the various components of magnetization identified in the Wealden redbeds (except for group 4 of recent origin). 1. Upper Jurassic–lower Cretaceous, 2. Upper Cretaceous. 3. Eocene: (a) mean directions and confidence limits of groups 1–3; (b) reference directions and confidence limits (North America for upper Jurassic–lower Cretaceous, Europe for upper Cretaceous and lower Tertiary).

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